go-On — Faculty & Class Advisor Timetable Matrix Integration

System Architecture, Reusable Grid Implementation, Layout Responsiveness & Release Audit

Target Device: Samsung Galaxy S24 (393 × 851 dp)	**Reference Design:** Student Weekly Matrix Panel
Release Build: 66.2 MB (app-release.apk)	**Static Analysis:** 0 Issues (flutter analyze)
Timetable Grid: 7 Days × 9 Slots (P1-P7 + 2 Breaks)	**Theme Compatibility:** Warm Light & Obsidian Dark

1. Executive Summary & Objective

The user requested replicating the student portal's weekly matrix timetable panel across both the **Faculty Portal** and the **Class Advisor Portal**. The reference design features a 7-day column header (Monday through Sunday), a vertical time rail indicating start times for all 9 daily slots (P1–P7 plus institutional Tea and Lunch breaks), pastel-toned subject cards, automated merging of consecutive periods (such as lab sessions), and an interactive bottom sheet for detailed class exploration.

The solution was engineered with full architectural parity across both portals: 1) A reusable WeeklyMatrixTimetableView component that renders across all screen densities without overflow; 2) A view mode toggle (Day Agenda vs. Weekly Grid) and advisor scope selector (My Schedule vs. Class Schedule) in the Faculty Dashboard; 3) A 4th tab (CLASS SCHEDULE) in the Class Advisor Portal; and 4) High-performance backend endpoints serving complete 42-entry weekly timetables.

2. Technical Architecture & Component Structure

Component	File Path	Key Functional Role
FastAPI Endpoints	`app/api/faculty.py`	Serves full weekly schedule for advisor's assigned class (42 periods) & individual class lookups.
API Config & Models	`api_config.dart` `faculty_model.dart`	Configures endpoints and maps FacultyScheduleItem.toTimetableItemModel().
Matrix Widget	`weekly_matrix_timetable_view.dart`	Reusable 7-day × 9-slot matrix with time rail, break dividers, pastel styling & consecutive merging.
Faculty Dashboard	`faculty_dashboard_screen.dart`	Adds View Mode toggle (Day Agenda / Weekly Grid) and Advisor Scope toggle (My / Class Schedule).
Advisor Portal	`faculty_class_advisor_screen.dart`	Adds 4th tab: CLASS SCHEDULE with responsive scrollable tab bar and real-time data.

3. Visual Design & S24 Responsive Calibration

To ensure pixel-perfect fidelity on the target Samsung Galaxy S24 (393 × 851 dp), the following layout rules were enforced:
• **Synchronized 2D Scrolling:** The left time rail remains pinned while the 7 weekday columns scroll smoothly horizontally.
• **Dynamic Pastel Coding:** Distinct soft pastel tints matching the reference screenshot (#EAE3F7 FDS Lavender, #FDE7D2 DMC Peach, #D9F2E6 OOPS Mint, #DCEBFB DPCO Soft Blue, #FEE0E0 DS Rose, #FFF3D6 IKS Light Amber).
• **Non-Teaching Break Bars:** Institutional breaks (Tea Break 10:35–10:55, Lunch Break 12:45–01:45) render as clear divider bars rather than selectable teaching cards, preserving attendance denomination rules.
• **Zero RenderFlex Overflow:** Flexible text wrappers and constrained title widths eliminate any truncation or overflow errors in both Warm Light and Obsidian Dark themes.

4. Automated Verification & Quality Assurance

Test Suite	Target Scope	Result
`faculty_and_advisor_timetable_matrix_test.dart`	Matrix component rendering, view toggles, scope switching, S24 dark theme	**PASSED** (4/4)
`faculty_timetable_redesign_test.dart`	Regression: Day capsule selector, Monday session cards, Sunday empty state	**PASSED** (7/7)
`faculty_class_advisor_lunch_break_removal_test.dart`	Regression: Lunch break exclusion from attendance denominators & UI	**PASSED** (2/2)
`student_timetable_seven_day_test.dart`	Regression: 7-day student timetable integrity & calendar navigation	**PASSED** (5/5)
`flutter analyze`	Static analysis, type checking, lint rules, import sanitization	**0 ISSUES**

5. Deployment & Release Artifacts

The production release binary has been built and verified:
• **Binary Path:** E:\originalam\app-release.apk (Size: 66.2 MB compiled / 69.3 MB on disk)
• **Distribution Server:** Active on Port 8080 (scripts/serve_apk.py)
• **Direct Download Link:** http://192.168.29.56:8080/app-release.apk
• **FastAPI Backend:** Active on Port 8000 serving timetable APIs.